NEET UG HIGH-YIELD PHYSICS

Atomic Structure and Hydrogen Spectra

Complete Chapter Master Tutorial: From Thomson and Rutherford Models to Bohr's Quantum Postulates and de Broglie Verification

1. Syllabus Overview and Learning Objectives

In NEET UG, the topic of **Atoms** consistently contributes 1 to 2 high-scoring questions every year. Modern Physics questions are formula-driven, conceptually rigorous, and calculation-sensitive. Mastering this module requires not only memorizing expressions for radii, orbital speeds, and energy states, but also appreciating the historical progression, Coulomb scattering mechanics, and quantum orbital transitions.

Core Learning Goals for NEET UG:

- Understand the qualitative and quantitative features of the **Geiger-Marsden α-particle scattering experiment**.
- Derive and calculate the **distance of closest approach** (d) and relate the **impact parameter** (b) to scattering angle (θ).
- Examine the classical **Rutherford planetary model** and recognize its mechanical and electrodynamic failures (the spiraling electron catastrophe and continuous spectra).
- Master **Bohr's three postulates** and derive the proportionality relations for orbital radius ($r_n \propto n^2/Z$), velocity ($v_n \propto Z/n$), and energy ($E_n \propto -Z^2/n^2$).
- Calculate photon frequencies, wavelengths, excitation potentials, and ionization energies across the **Lyman, Balmer, Paschen, Brackett, and Pfund** spectral series.
- Explain Bohr's second postulate of angular momentum quantization using **de Broglie's standing matter waves**.

2. Early Atomic Models and Rutherford's Scattering Experiment

2.1 Thomson's Model and Its Limitations

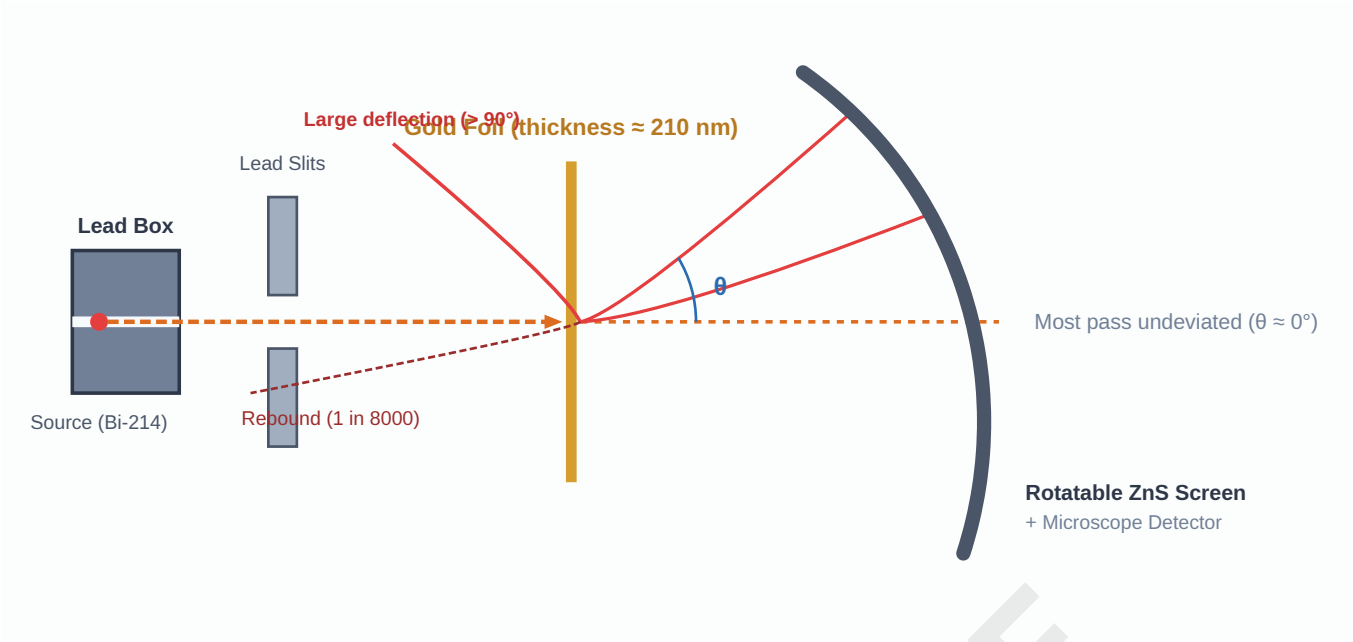
Following J. J. Thomson's 1897 discovery of the electron, he proposed in 1898 the *plum pudding model* (or watermelon model). The atom was pictured as a uniform spherical cloud of positive charge of diameter approximately $10^{-10}\ \text{m}$, within which negatively charged electrons were embedded in stable electrostatic equilibrium.

However, Thomson's model could not account for large-angle particle deflections, nor could it explain the sharp, discrete emission line spectra observed in rarefied atomic gases.

2.2 The Geiger-Marsden α-Particle Scattering Experiment (1911)

At the suggestion of Ernst Rutherford, Hans Geiger and Ernst Marsden directed a collimated beam of $5.5\ \text{MeV}$ α-particles (helium nuclei, $q = +2e$, mass $m_\alpha \approx 4\,u$) emitted from a radioactive bismuth source ($^{214}_{83}\text{Bi}$) onto an extremely thin gold foil of thickness $2.1 \times 10^{-7}\ \text{m}$.

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Tutorial Diagram 1: Schematic of the Geiger-Marsden α-particle scattering apparatus inside an evacuated chamber.

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2.3 Experimental Observations

1. **Dominant Transmission:** More than 99.8% of the incident α-particles passed directly through the gold foil with little or no deflection ($\theta < 1^\circ$).
2. **Small-Angle Deflection:** Only about 0.14% of the incident α-particles scattered by an angle greater than 1°.
3. **Extreme Backward Scattering:** Only about 1 in 8000 α-particles deflected by more than 90°, and a minuscule fraction suffered head-on rebound ($\theta \approx 180^\circ$).

Rutherford's Famous Deduction:

Rutherford remarked: *"It was quite the most incredible event that has ever happened to me in my life. It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you."*

Because electrons are extremely light ($m_e \approx m_\alpha/7300$), they cannot significantly deflect a fast-moving, massive α-particle. Deflection through large angles requires a concentrated, massive, positively charged core: the **atomic nucleus**.

2.4 Theoretical Distribution of Scattered Particles

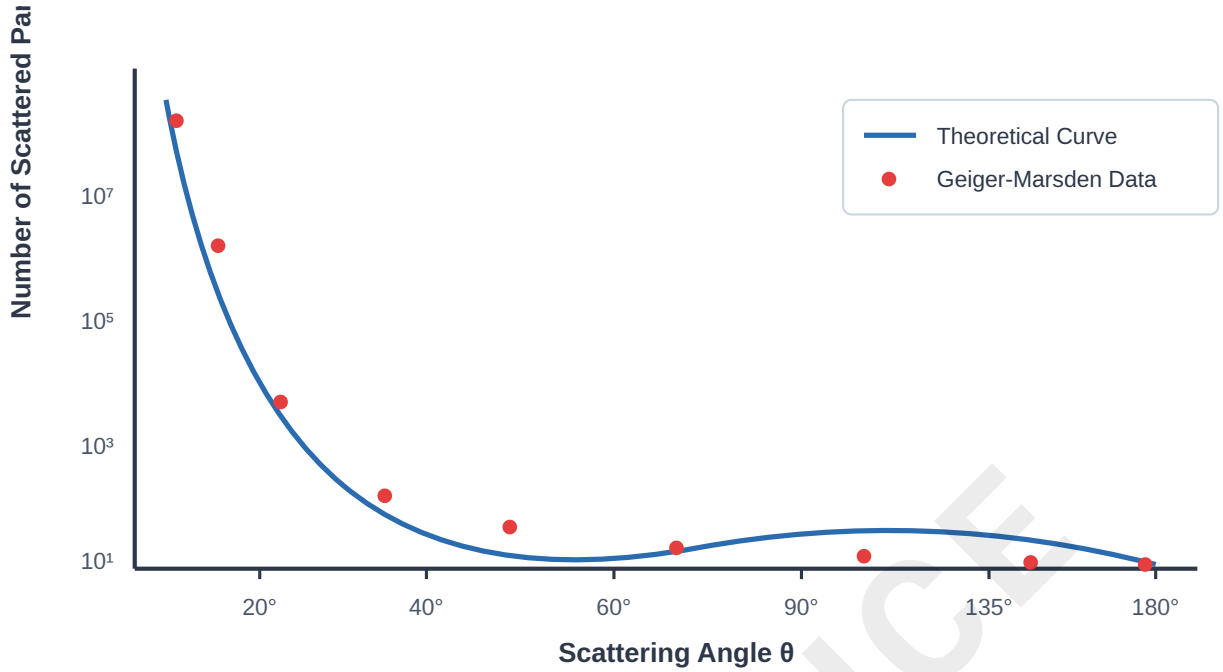
Assuming single-event elastic scattering governed by Coulomb's repulsive law between the target nucleus (charge $+Ze$) and the α-particle (charge $+2e$):

$$F = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{r^2}$$

Rutherford derived the relationship for the number of scattered particles $N(\theta)$ detected per unit area at scattering angle θ:

$$N(\theta) \propto \frac{1}{\sin^4(\theta/2)}$$

This inverse fourth-power dependence was verified precisely by Geiger and Marsden's experimental data points, proving that the Coulomb inverse-square law holds true down to nuclear distances.

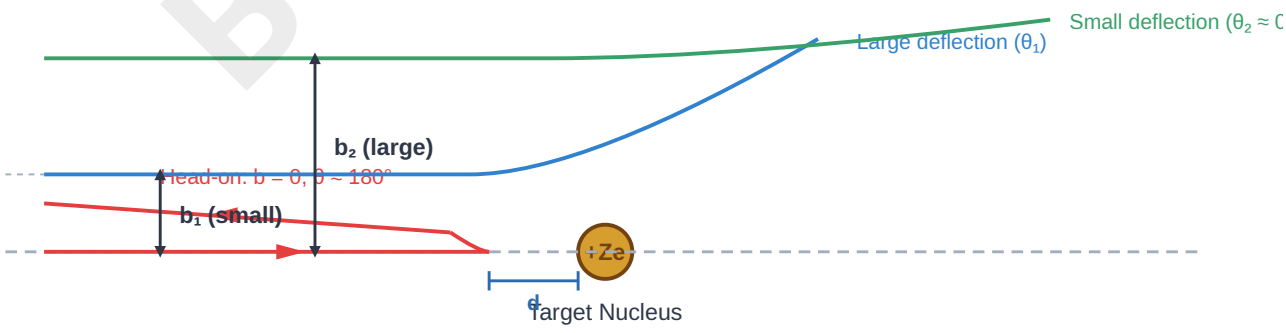


Tutorial Diagram 2: Number of scattered α-particles versus scattering angle θ. Experimental data strictly follows the theoretical curve $N \propto \csc^4(\theta/2)$.

3. Impact Parameter and Distance of Closest Approach

3.1 Impact Parameter (b)

The **impact parameter** b is defined as the perpendicular distance of the initial velocity vector of the incident α-particle from the center of the target nucleus.



Tutorial Diagram 3: Trajectories of α-particles in the Coulomb field of a gold nucleus for various impact parameters b.

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$$b = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2\cot(\theta/2)}{K}$$

where $K = \frac{1}{2}m_\alpha v^2$ is the kinetic energy of the incoming α-particle and θ is the scattering angle.

Physical Deductions:

- For **large impact parameters** ($b \to \infty$): $\cot(\theta/2) \to \infty \implies \theta \to 0$. The particle feels negligible force and passes undeviated.
- For **head-on collision** ($b = 0$): $\cot(\theta/2) = 0 \implies \theta = 180^\circ$ (π). The particle momentarily halts and rebounds along its incident line.

3.2 Distance of Closest Approach (d)

In a direct head-on collision ($b = 0$), the α-particle travels straight toward the nucleus. As it approaches, its initial kinetic energy K converts entirely into electrostatic potential energy U. At distance d, the particle momentarily comes to rest ($v = 0$) before reversing its velocity.

$$K = U(d) = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{d} = \frac{2Ze^2}{4\pi\varepsilon_0 d}$$

$$d = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{K}$$

This simple formula provides an upper limit for the size of the atomic nucleus.

Worked Example 1: Upper Limit to Gold Nucleus Radius

Problem: Calculate the distance of closest approach for a $7.7\ \text{MeV}$ α-particle directed head-on toward a gold nucleus ($Z = 79$). Given $\frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2$ and $e = 1.6 \times 10^{-19}\ \text{C}$.

Solution:

Convert the kinetic energy into Joules:

$$K = 7.7 \times 10^6 \times 1.6 \times 10^{-19}\ \text{J} \approx 1.23 \times 10^{-12}\ \text{J}$$

Substitute into the closest approach equation:

$$d = \frac{(9.0 \times 10^9) \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{1.23 \times 10^{-12}}$$

$$d = \frac{(9.0 \times 10^9) \times 158 \times 2.56 \times 10^{-38}}{1.23 \times 10^{-12}} \approx 2.96 \times 10^{-14}\ \text{m} \approx 30\ \text{fm}$$

Conclusion: The actual physical radius of a gold nucleus determined by high-energy electron scattering is approximately $6\ \text{fm}$. Because $d \approx 30\ \text{fm} > 6\ \text{fm}$, the α-particle turns back purely under electrostatic repulsion without ever coming into contact with the nuclear surface.

4. Rutherford's Planetary Model and Its Inherent Failure

4.1 Mechanical Equilibrium of Electron Orbits

Rutherford envisioned an atom with a central positive nucleus surrounded by orbiting electrons, balanced by the electrostatic Coulomb attractive force acting as the centripetal force:

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$$F_e = F_c \implies \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2} = \frac{mv^2}{r}$$

Hence, the orbital radius is related to electron speed by:

$$r = \frac{e^2}{4\pi\varepsilon_0 m v^2}$$

4.2 Orbital Energy Relations

- **Kinetic Energy (K):**

$$K = \frac{1}{2}mv^2 = \frac{e^2}{8\pi\varepsilon_0 r}$$

- **Electrostatic Potential Energy (U):**

$$U = -\frac{e^2}{4\pi\varepsilon_0 r}$$

- **Total Mechanical Energy (E):**

$$E = K + U = \frac{e^2}{8\pi\varepsilon_0 r} - \frac{e^2}{4\pi\varepsilon_0 r} = -\frac{e^2}{8\pi\varepsilon_0 r}$$

Vital NEET Energy Relation:

$$E = -K = \frac{U}{2} \quad \Longleftrightarrow \quad U = 2E = -2K$$

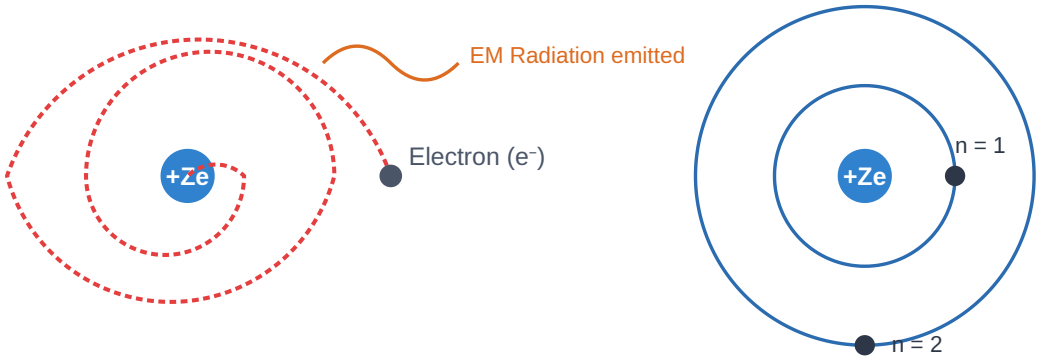
The negative sign of E confirms that the electron forms a *bound system* with the nucleus.

4.3 Classical Electrodynamic Collapse and Spectrum Contradiction

Classical physics (Maxwell's electrodynamics) makes two clear predictions that directly contradict experimental observations of real atoms:

1. **Atomic Instability (Spiraling Electron):** An orbiting electron experiences continuous centripetal acceleration ($a = v^2/r$). Classical electromagnetic theory dictates that any accelerating charged particle must continuously radiate energy as electromagnetic waves. As it loses energy, its total energy becomes more negative, reducing its orbit radius. Calculations show the electron would spiral into the nucleus in approximately 10^{-8} s, making stable matter impossible.
2. **Continuous Emission Spectrum:** In classical theory, the frequency of radiation emitted by a revolving electron equals its orbital frequency of revolution ($\nu_{\text{rev}} = v/2\pi r$). As the electron spirals inward and r shrinks, ν_{rev} would increase continuously, producing a continuous rainbow spectrum instead of the discrete, isolated bright spectral lines actually observed.

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Classical Prediction: Collapse in ~10^{-8} s

Bohr's Postulate: Non-Radiating Stable States

Tutorial Diagram 4: Contrast between Rutherford's classical radiative orbital collapse and Bohr's non-radiating discrete stationary orbits.

5. The Bohr Model of the Hydrogen Atom

In 1913, Niels Bohr resolved the stability and spectral paradoxes by combining classical Coulomb mechanics with early quantum principles (Planck and Einstein). His model rests on three fundamental postulates.

5.1 Bohr's Three Postulates

Postulate 1: Stationary Non-Radiating States

An electron in an atom can revolve in certain specific circular orbits without radiating electromagnetic energy, contrary to classical electrodynamics. Each allowed orbit has a fixed, discrete energy value and is called a **stationary state**.

Postulate 2: Angular Momentum Quantization

The orbital angular momentum (L) of an electron revolving around the nucleus is an integral multiple of $h/2\pi$:

$$L = mvr = \frac{nh}{2\pi}, \quad \text{where } n = 1, 2, 3, \ldots \text{ (principal quantum number)}$$

Postulate 3: Frequency Condition (Photon Transition)

Radiation is emitted or absorbed only when an electron jumps from one non-radiating orbit to another. A single photon is emitted when dropping from higher state E_i to lower state E_f:

$$h\nu = E_i - E_f \implies \nu = \frac{E_i - E_f}{h}$$

5.2 Mathematical Derivations for Hydrogen-Like Species (Z)

Consider a single electron of mass m and charge $-e$ revolving at speed v in an orbit of radius r around a nucleus of positive charge $+Ze$.

1. Radius of the nth Orbit:

From the dynamic balance:

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$$\frac{mv^2}{r} = \frac{1}{4\pi\varepsilon_0}\frac{Ze^2}{r^2} \implies v^2 = \frac{Ze^2}{4\pi\varepsilon_0 mr}$$

From the quantization condition: $v = \frac{nh}{2\pi mr}$. Squaring this gives:

$$\frac{n^2h^2}{4\pi^2m^2r^2} = \frac{Ze^2}{4\pi\varepsilon_0 mr}$$

Solving for r_n:

$$r_n = \frac{\varepsilon_0 h^2}{\pi m e^2}\cdot\frac{n^2}{Z} = a_0\frac{n^2}{Z}$$

For hydrogen ($Z=1, n=1$), the Bohr radius is $a_0 = \frac{\varepsilon_0 h^2}{\pi m e^2} \approx 0.529\times10^{-10}\ \text{m} = 0.529\ \text{Å}$.

2. Orbital Speed in the nth Orbit:

Substituting r_n into $v = nh/(2\pi mr)$:

$$v_n = \frac{e^2}{2\varepsilon_0 h}\cdot\frac{Z}{n} = v_1\frac{Z}{n}$$

For hydrogen ($Z=1, n=1$), $v_1 \approx 2.18\times10^6\ \text{m/s} \approx \frac{c}{137}$.

3. Orbital Period (T) and Revolution Frequency (f):

$$T_n = \frac{2\pi r_n}{v_n} \propto \frac{n^2/Z}{Z/n} \propto \frac{n^3}{Z^2}$$

$$f_n = \frac{1}{T_n} \propto \frac{Z^2}{n^3}$$

4. Total Energy of the Stationary State:

$$E_n = -\frac{Ze^2}{8\pi\varepsilon_0 r_n} = -\left(\frac{me^4}{8\varepsilon_0^2h^2}\right)\frac{Z^2}{n^2}$$

Substituting the values of fundamental constants (m, e, ε_0, h):

$$E_n = -\frac{2.18\times10^{-18}\ \text{J}}{n^2}\cdot Z^2 = -13.6\frac{Z^2}{n^2}\ \text{eV}$$

5.3 Hydrogen Energy Levels and State Terminology

Principal Quantum No. (n)	State Designation	Energy (E_n)	Excitation Energy from Ground State	Binding / Ionization Energy
$n=1$	Ground State	-13.60 eV	0.00 eV	13.60 eV
$n=2$	1st Excited State	-3.40 eV	10.20 eV	3.40 eV
$n=3$	2nd Excited State	-1.51 eV	12.09 eV	1.51 eV
$n=4$	3rd Excited State	-0.85 eV	12.75 eV	0.85 eV
$n=\infty$	Ionized Atom / Free Electron	0.00 eV	13.60 eV	0.00 eV

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NEET Traps in Terminology:

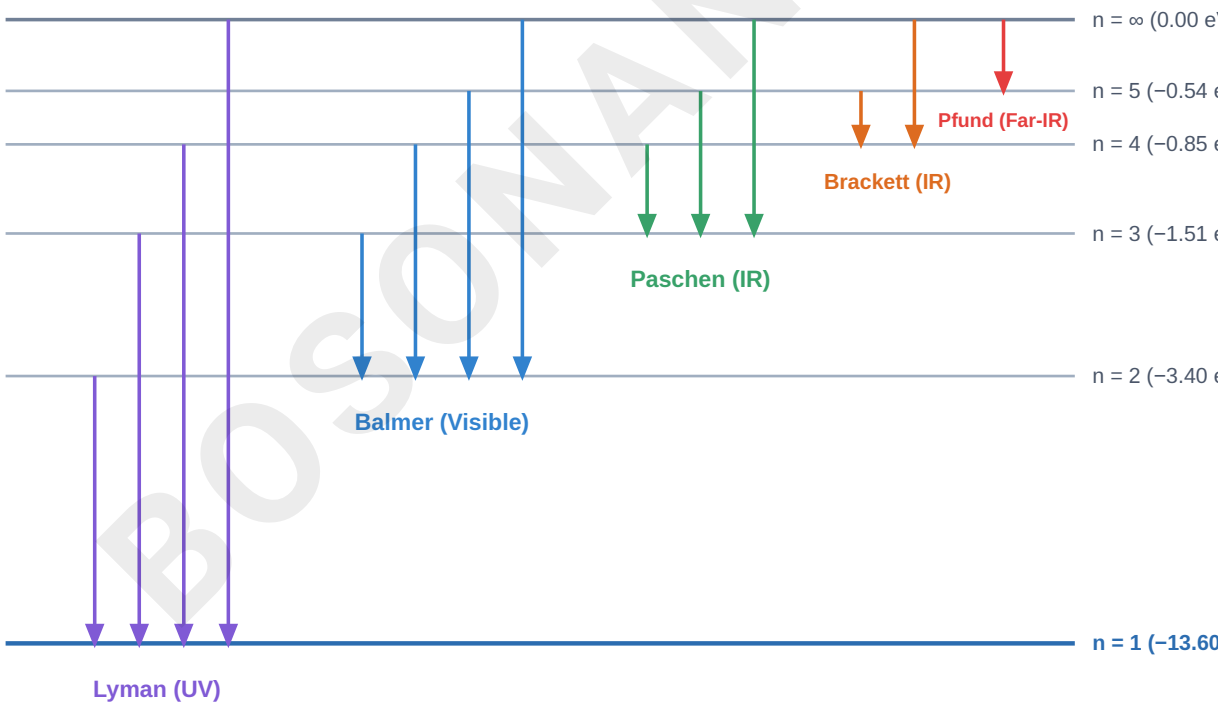
- **"First excited state"** means $n = 2$, NOT $n = 1$.
- **"m^{th} excited state"** corresponds to quantum number $n = m + 1$.
- **Ionization Potential** of ground-state hydrogen is $13.6\ \text{V}$.

6. Spectral Series of Hydrogen and Rydberg Formula

When an electron makes a transition from an outer orbit n_i to an inner orbit n_f ($n_i > n_f$), a photon is emitted. The wavenumber $\bar{\nu}$ (reciprocal of wavelength λ) is given by the **Rydberg-Ritz formula**:

$$\frac{1}{\lambda} = \bar{\nu} = RZ^2 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

where $R = \frac{me^4}{8\varepsilon_0^2 ch^3} \approx 1.097 \times 10^7\ \text{m}^{-1}$. For rapid calculations in NEET: $\frac{1}{R} \approx 912$ Å.



Tutorial Diagram 5: Energy level diagram of atomic hydrogen showing radiative transitions for each spectral series.

6.1 Summary of Hydrogen Spectral Series

Series	n_f	n_i	Spectral Region	Longest Wavelength (λ_{max})	Shortest Wavelength (Series Limit λ_{min})
Lyman	1	$2, 3, 4, \ldots$	Ultraviolet (UV)	$n_i = 2 \implies \frac{4}{3R} \approx 1216$ Å	$n_i = \infty \implies \frac{1}{R} \approx 912$ Å
Balmer	2	$3, 4, 5, \ldots$	Visible & Near-UV	$n_i = 3 \implies \frac{36}{5R} \approx 6563$ Å (H_α)	$n_i = \infty \implies \frac{4}{R} \approx 3646$ Å

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Series	n_f	n_i	Spectral Region	Longest Wavelength ($\lambda_{\max}$)	Shortest Wavelength (Series Limit $\lambda_{\min}$)
Paschen	3	$4, 5, 6, \ldots$	Infrared (Near-IR)	$n_i = 4 \implies \frac{144}{7R} \approx 18751$ Å	$n_i = \infty \implies \frac{9}{R} \approx 8205$ Å
Brackett	4	$5, 6, 7, \ldots$	Infrared (Mid-IR)	$n_i = 5 \implies \frac{400}{9R} \approx 40510$ Å	$n_i = \infty \implies \frac{16}{R} \approx 14585$ Å
Pfund	5	$6, 7, 8, \ldots$	Far Infrared	$n_i = 6 \implies \frac{900}{11R} \approx 74580$ Å	$n_i = \infty \implies \frac{25}{R} \approx 22790$ Å

Maximum Spectral Lines Emitted:

When an electron in an excited state n drops toward the ground state ($n = 1$), the total number of distinct spectral emission lines emitted by a sample of gas is:

$$N = \frac{n(n-1)}{2}$$

For de-excitation between arbitrary levels $n_2 \to n_1$:

$$N = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

7. De Broglie's Explanation of Bohr's Quantization Condition

Bohr's second postulate ($mvr = nh/2\pi$) appeared as an ad-hoc rule until Louis de Broglie provided its physical foundation in 1923 using wave-particle duality.

De Broglie postulated that a moving electron behaves as a matter wave with de Broglie wavelength:

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

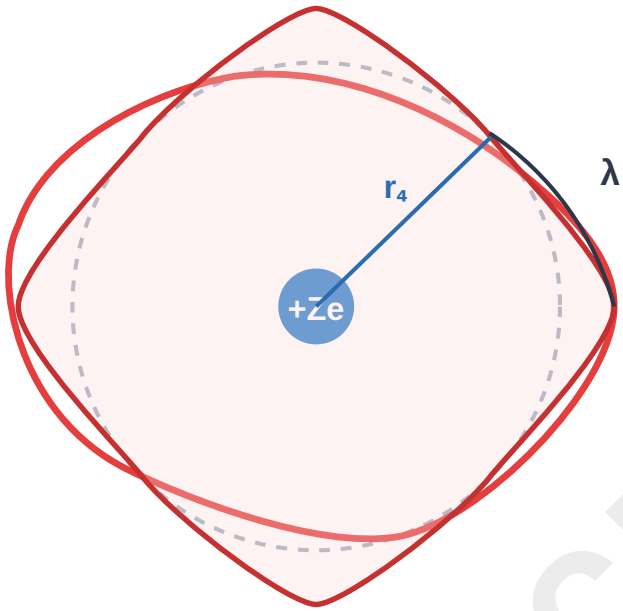
For an electron to form a stable, non-radiating standing wave on a circular orbit of radius r_n, the total perimeter of the orbit must fit an integral number of complete de Broglie wavelengths without destructive self-interference:

$$2\pi r_n = n\lambda = n\left(\frac{h}{mv_n}\right), \quad n = 1, 2, 3, \ldots$$

Rearranging immediately yields Bohr's quantization condition:

$$mv_n r_n = \frac{nh}{2\pi}$$

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Standing Wave on Orbit: $2\pi r_4 = 4\lambda$ (n = 4)

Tutorial Diagram 6: Standing de Broglie matter wave on a circular orbit for $n = 4$. The circumference holds exactly 4 wavelengths.

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8. Limitations of the Bohr Model

While Bohr's model achieved great success for hydrogen, it has distinct limitations that necessitated modern quantum mechanics:

- **Limited to Hydrogenic Species:** It applies only to single-electron systems ($\mathrm{H}, \mathrm{He}^{+}, \mathrm{Li}^{2+}, \mathrm{Be}^{3+}$). It fails for two-electron helium (He) or multi-electron atoms because it neglects electron-electron electrostatic repulsion.
- **Relative Intensities:** It cannot explain why certain spectral lines appear brighter (stronger intensity) than others (some transitions have higher quantum transition probabilities).
- **Fine Structure:** High-resolution spectrometers show that individual spectral lines are actually closely spaced doublets or triplets, which Bohr's simple model cannot predict.
- **Zeeman and Stark Effects:** It fails to explain the splitting of spectral lines in an external magnetic field (**Zeeman effect**) or an external electric field (**Stark effect**).
- **Violation of Uncertainty Principle:** Bohr assumes a well-defined circular orbit with simultaneous exact knowledge of the electron's position and momentum, which contradicts the **Heisenberg Uncertainty Principle**.

9. Master Formula Sheet for NEET UG

Physical Parameter	General Formula (Hydrogenic Species Z)	Dependence on n and Z	Value for Hydrogen ($n = 1, Z = 1$)
Orbital Radius (r_n)	$r_n = \frac{\varepsilon_0 h^2 n^2}{\pi m Z e^2}$	$r_n \propto \frac{n^2}{Z}$	$a_0 = 0.529\ \text{Å}$
Orbital Speed (v_n)	$v_n = \frac{Z e^2}{2\varepsilon_0 n h}$	$v_n \propto \frac{Z}{n}$	$v_1 = 2.18 \times 10^6\ \text{m/s}$

Physical Parameter	General Formula (Hydrogenic Species Z)	Dependence on n and Z	Value for Hydrogen ($n=1, Z=1$)
Time Period (T_n)	$T_n = \frac{2\pi r_n}{v_n}$	$T_n \propto \frac{n^3}{Z^2}$	$T_1 \approx 1.52 \times 10^{-16}\ \text{s}$
Orbital Frequency (f_n)	$f_n = \frac{1}{T_n}$	$f_n \propto \frac{Z^2}{n^3}$	$f_1 \approx 6.6 \times 10^{15}\ \text{Hz}$
Kinetic Energy (K_n)	$K_n = \frac{me^4Z^2}{8\varepsilon_0^2 h^2 n^2}$	$K_n \propto \frac{Z^2}{n^2}$	$+13.6\ \text{eV}$
Potential Energy (U_n)	$U_n = -2K_n = -\frac{me^4Z^2}{4\varepsilon_0^2 h^2 n^2}$	$U_n \propto -\frac{Z^2}{n^2}$	$-27.2\ \text{eV}$
Total Energy (E_n)	$E_n = -K_n = -\frac{13.6Z^2}{n^2}\ \text{eV}$	$E_n \propto -\frac{Z^2}{n^2}$	$-13.6\ \text{eV}$
Angular Momentum (L_n)	$L_n = \frac{nh}{2\pi}$	$L_n \propto n$ (independent of Z)	$\hbar = 1.055 \times 10^{-34}\ \text{J}\cdot\text{s}$
Closest Approach (d)	$d = \frac{2Ze^2}{4\pi\varepsilon_0 K}$	$d \propto \frac{Z}{K}$	$\approx 30\ \text{fm}$ (for $7.7\ \text{MeV}$ on Au)

10. Step-by-Step Solved Textbook Examples

Worked Example 2: Solar System vs. Atomic Empty Space Proportion

Problem: In the nuclear atom model, the nucleus has radius $\approx 10^{-15}\ \text{m}$ while the atom has radius $\approx 10^{-10}\ \text{m}$. If the solar system had the same proportions as an atom, calculate whether Earth would be closer to or farther from the Sun than it actually is. The Sun's radius is $7 \times 10^8\ \text{m}$ and Earth's actual orbit radius is $1.5 \times 10^{11}\ \text{m}$.

Solution:

1. Ratio of atomic radius to nuclear radius:

$$\frac{R_{\text{atom}}}{R_{\text{nucleus}}} = \frac{10^{-10}\ \text{m}}{10^{-15}\ \text{m}} = 10^5$$

2. Scaled orbit radius of Earth:

$$R_{\text{scaled}} = 10^5 \times R_{\text{Sun}} = 10^5 \times (7 \times 10^8\ \text{m}) = 7 \times 10^{13}\ \text{m}$$

3. Compare with actual orbital radius ($1.5 \times 10^{11}\ \text{m}$):

$$\frac{R_{\text{scaled}}}{R_{\text{actual}}} = \frac{7 \times 10^{13}}{1.5 \times 10^{11}} \approx 467$$

Answer: Earth would be over 400 times farther away from the Sun than it currently is. This demonstrates that an atom contains a vastly larger fraction of empty space than our solar system.

Worked Example 3: Classical Initial Frequency of Radiation

Problem: According to classical electromagnetic theory, calculate the initial frequency of light emitted by an electron revolving around a proton in a hydrogen atom in an orbit of radius $5.3 \times 10^{-11}\ \text{m}$ at a speed of $2.2 \times 10^6\ \text{m/s}$.

Solution:

In classical electrodynamics, the radiated wave frequency equals the orbital revolution frequency:

$$\nu_{\text{classical}} = f_{\text{rev}} = \frac{v}{2\pi r} = \frac{2.2 \times 10^6}{2 \times 3.1416 \times (5.3 \times 10^{-11})}$$

$$\nu_{\text{classical}} = \frac{2.2 \times 10^6}{3.33 \times 10^{-10}} \approx 6.6 \times 10^{15}\ \text{Hz}$$

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Answer: The initial frequency of classically predicted emitted radiation is $6.6 \times 10^{15}\ \text{Hz}$ (which falls in the ultraviolet region).

Worked Example 4: Photon Emission upon Inelastic Electron Bombardment

Problem: A beam of $12.5\ \text{eV}$ electrons bombards gaseous hydrogen at room temperature (initially in the ground state). Which spectral series and lines will be emitted when the atoms de-excite?

Solution:

1. Energy required to excite from $n = 1$ ($-13.6\ \text{eV}$) to higher states:
   - To $n = 2$: $\Delta E_{1\to2} = -3.4 - (-13.6) = 10.2\ \text{eV}$
   - To $n = 3$: $\Delta E_{1\to3} = -1.51 - (-13.6) = 12.09\ \text{eV}$
   - To $n = 4$: $\Delta E_{1\to4} = -0.85 - (-13.6) = 12.75\ \text{eV}$

2. Because the incident beam energy is $12.5\ \text{eV}$, it is sufficient to excite electrons to $n = 3$ ($12.09\ \text{eV} < 12.5\ \text{eV}$), but insufficient to reach $n = 4$ ($12.75\ \text{eV} > 12.5\ \text{eV}$).

3. Possible de-excitation pathways from $n = 3$:
   - $n = 3 \to n = 1$: Lyman series (L_β line, $\lambda \approx 102.6\ \text{nm}$)
   - $n = 2 \to n = 1$: Lyman series (L_α line, $\lambda \approx 121.6\ \text{nm}$)
   - $n = 3 \to n = 2$: Balmer series (H_α line, $\lambda \approx 656.3\ \text{nm}$)

Answer: The emitted radiation consists of 3 distinct lines: two lines in the Lyman series (ultraviolet) and one line in the Balmer series (visible red).

11. Original NEET UG Practice Questions

Practice Question 1

Single Option Correct (NEET Standard)

In a Geiger-Marsden scattering experiment, an α-particle of kinetic energy K collides head-on with a gold nucleus ($Z = 79$) and stops momentarily at a distance of closest approach d. If the kinetic energy of the incident α-particle is doubled, the new distance of closest approach will become:

(A) $2d$

(B) $\sqrt{2}d$

(C) $\frac{d}{2}$

(D) $\frac{d}{4}$

Correct Answer: (C)

Explanation: In a head-on collision, initial kinetic energy converts entirely into electrostatic potential energy:

$$K = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{d} \implies d \propto \frac{1}{K}$$

When kinetic energy is doubled ($K' = 2K$), the distance of closest approach halves: $d' = \frac{d}{2}$.

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Practice Question 2

Scattering Angle Dependence

At a scattering angle of $\theta = 60^\circ$, the number of α-particles scattered per unit area is observed to be N. According to Rutherford's scattering formula, what will be the number of α-particles scattered per unit area at an angle of $\theta = 90^\circ$?

(A) $\frac{N}{4}$

(B) $\frac{N}{2}$

(C) $2N$

(D) $4N$

Correct Answer: (A)

Explanation: The Rutherford scattering law states:

$$N(\theta) \propto \frac{1}{\sin^4(\theta/2)}$$

For $\theta_1 = 60^\circ$: $\sin(\theta_1/2) = \sin(30^\circ) = \frac{1}{2} \implies \sin^4(30^\circ) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$.

For $\theta_2 = 90^\circ$: $\sin(\theta_2/2) = \sin(45^\circ) = \frac{1}{\sqrt{2}} \implies \sin^4(45^\circ) = \left(\frac{1}{\sqrt{2}}\right)^4 = \frac{1}{4}$.

Taking the ratio:

$$\frac{N(90^\circ)}{N(60^\circ)} = \frac{\sin^4(30^\circ)}{\sin^4(45^\circ)} = \frac{1/16}{1/4} = \frac{1}{4} \implies N(90^\circ) = \frac{N}{4}$$

Practice Question 3

Bohr Orbit Proportionalities

If the radius of the first orbit of a hydrogen atom ($n = 1$) is a_0, then the radius of the second orbit of a singly ionized helium atom (He^+, $Z = 2$) is:

(A) a_0

(B) $2a_0$

(C) $4a_0$

(D) $0.5a_0$

Correct Answer: (B)

Explanation: The radius of the nth orbit in a hydrogen-like atom is:

$$r_n = a_0 \frac{n^2}{Z}$$

For He^+, $Z = 2$. For the second orbit, $n = 2$:

$$r_2 = a_0 \frac{(2)^2}{2} = a_0 \frac{4}{2} = 2a_0$$

BOSONANCE

Practice Question 4

Orbital Period Scaling

The period of revolution of an electron in the ground state of a hydrogen atom is T. The period of revolution of the electron in the first excited state ($n = 2$) will be:

(A) $2T$

(B) $4T$

(C) $8T$

(D) $16T$

Correct Answer: (C)

Explanation: Orbital period is related to principal quantum number by:

$$T \propto \frac{r}{v} \propto \frac{n^2}{1/n} = n^3$$

For the first excited state, $n = 2$. Hence:

$$T_2 = T_1 \times (2)^3 = 8T$$

Practice Question 5

Potential vs Total Energy

The total energy of an electron in a specific stationary orbit of a hydrogen atom is $-3.4\ \text{eV}$. Its kinetic energy (K) and electrostatic potential energy (U) in this orbit are respectively:

(A) $+3.4\ \text{eV}$ and $-6.8\ \text{eV}$

(B) $-3.4\ \text{eV}$ and $-6.8\ \text{eV}$

(C) $+6.8\ \text{eV}$ and $-3.4\ \text{eV}$

(D) $+3.4\ \text{eV}$ and $+6.8\ \text{eV}$

Correct Answer: (A)

Explanation: In Bohr's orbit:

$$K = -E = -(-3.4\ \text{eV}) = +3.4\ \text{eV}$$
$$U = 2E = 2 \times (-3.4\ \text{eV}) = -6.8\ \text{eV}$$

BOSONANCE

Practice Question 6

Spectral Series Wavelength Ratio

The ratio of the longest wavelength in the Lyman series to the longest wavelength in the Balmer series of the hydrogen spectrum is:

(A) $\frac{5}{27}$

(B) $\frac{27}{5}$

(C) $\frac{4}{9}$

(D) $\frac{5}{36}$

Correct Answer: (A)

Explanation: Longest wavelength corresponds to the minimum energy jump between adjacent levels:

1. Lyman longest wavelength ($2 \to 1$):

$$\frac{1}{\lambda_{L,\max}} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = R\left(1 - \frac{1}{4}\right) = \frac{3R}{4} \implies \lambda_{L,\max} = \frac{4}{3R}$$

2. Balmer longest wavelength ($3 \to 2$):

$$\frac{1}{\lambda_{B,\max}} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5R}{36} \implies \lambda_{B,\max} = \frac{36}{5R}$$

3. Ratio:

$$\frac{\lambda_{L,\max}}{\lambda_{B,\max}} = \frac{4/(3R)}{36/(5R)} = \frac{4}{3} \times \frac{5}{36} = \frac{20}{108} = \frac{5}{27}$$

Practice Question 7

Number of Spectral Lines

A sample of hydrogen atoms in their ground state absorbs photons such that the electrons are excited to the state with principal quantum number $n = 5$. How many different spectral emission lines can potentially be observed as the atoms return to the ground state?

(A) 4

(B) 6

(C) 10

(D) 15

Correct Answer: (C)

Explanation: The maximum number of spectral lines emitted during de-excitation from level n to ground state is:

$$N = \frac{n(n-1)}{2} = \frac{5(5-1)}{2} = \frac{5 \times 4}{2} = 10$$

BOSONANCE

Practice Question 8

De Broglie Standing Waves

According to de Broglie's hypothesis, the circumference of the third orbit ($n = 3$) of a hydrogen atom is:

(A) Equal to 3 de Broglie wavelengths of the electron in that orbit

(B) Equal to 1.5 de Broglie wavelengths of the electron in that orbit

(C) Equal to 9 de Broglie wavelengths of the electron in that orbit

(D) Independent of the de Broglie wavelength

Correct Answer: (A)

Explanation: The condition for resonant standing matter waves in an allowed circular orbit is:

$$2\pi r_n = n\lambda$$

For $n = 3$, the circumference $2\pi r_3 = 3\lambda$, meaning precisely 3 complete de Broglie wavelengths fit around the orbit.

Practice Question 9

Assertion and Reason (NEET Pattern)

Assertion (A): Thomson's atomic model could not explain the large-angle scattering of α-particles observed in the Geiger-Marsden experiment.
Reason (R): In Thomson's model, the positive charge and mass are uniformly distributed over the entire volume of the atom, resulting in weak electric fields that cannot produce large deflections.

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Explanation: In Thomson's model, the positive charge is diffuse over a sphere of radius $\sim 10^{-10}$ m. The maximum electric field produced by such a charge distribution is far too small to reverse or deflect a high-energy, massive α-particle. Only a concentrated nuclear core creates an electric field intense enough to cause large-angle deflections.

BOSONANCE

Practice Question 10

Statement-Based Question (NEET Pattern)

Evaluate the following two statements regarding atomic models:
Statement I: According to classical electromagnetic theory, an electron in Rutherford's planetary model must continuously emit radiation and spiral into the nucleus.
Statement II: Bohr's model is applicable to multi-electron systems such as neutral helium without any modifications.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct, but Statement II is incorrect.

(D) Statement I is incorrect, but Statement II is correct.

Correct Answer: (C)

Explanation: Statement I is correct: Maxwell's classical electrodynamics predicts that an accelerated charge continuously emits electromagnetic radiation, causing an orbital spiral. Statement II is incorrect: Bohr's model applies strictly to single-electron (hydrogenic) systems; it fails for neutral helium because it does not include electron-electron repulsion.

Practice Question 11

De-excitation and Transition Limits

The series limit wavelength for the Balmer series of hydrogen is λ_0. What is the series limit wavelength for the Lyman series of a singly ionized helium atom (He^+)?

(A) λ_0

(B) $\frac{\lambda_0}{4}$

(C) $\frac{\lambda_0}{16}$

(D) $4\lambda_0$

Correct Answer: (C)

Explanation: The series limit corresponds to a transition from $n_i = \infty$:
1. Balmer series limit of hydrogen ($Z = 1, n_f = 2$):

$$\frac{1}{\lambda_0} = R(1)^2 \left(\frac{1}{2^2} - 0\right) = \frac{R}{4} \implies \lambda_0 = \frac{4}{R}$$

2. Lyman series limit of He^+ ($Z = 2, n_f = 1$):

$$\frac{1}{\lambda_{\mathrm{He}^+}} = R(2)^2 \left(\frac{1}{1^2} - 0\right) = 4R \implies \lambda_{\mathrm{He}^+} = \frac{1}{4R}$$

3. Express λ_{He^+} in terms of λ_0:

$$\lambda_{\mathrm{He}^+} = \frac{1}{4R} = \frac{1}{16}\left(\frac{4}{R}\right) = \frac{\lambda_0}{16}$$

BOSONANCE

Practice Question 12

Angular Momentum Quantization

What is the change in orbital angular momentum of an electron when it makes a transition from the third excited state to the ground state in a hydrogen atom?

(A) $\frac{h}{2\pi}$

(B) $\frac{2h}{\pi}$

(C) $\frac{3h}{2\pi}$

(D) $\frac{3h}{\pi}$

Correct Answer: (C)

Explanation: "Third excited state" corresponds to $n = 3 + 1 = 4$. The ground state is $n = 1$.

$$\Delta L = L_4 - L_1 = \frac{4h}{2\pi} - \frac{1h}{2\pi} = \frac{3h}{2\pi}$$

12. Common Student Traps and High-Yield Revision Checklist

Frequent Conceptual and Numerical Errors in NEET:

- **State Numbering Confusion:** Ground state is $n = 1$. The nth excited state corresponds to principal quantum number $(n + 1)$. For example, the *second excited state* has $n = 3$.
- **Sign Conventions for Atomic Energy:** Total mechanical energy E and potential energy U are strictly negative for bound states; kinetic energy K is always positive. Remember: $K = -E$, $U = 2E$, and $K = -U/2$.
- **Target Atomic Number (Z) Dependency:** Closest approach distance $d \propto Z$, but radius of orbit $r \propto 1/Z$, velocity $v \propto Z$, and energy $E \propto Z^2$.
- **Spectral Series Wavelength Limits:** The longest wavelength (λ_{max}) occurs for transition between adjacent levels ($\Delta n = 1$). The shortest wavelength (λ_{min}, series limit) occurs for transition from $n_i = \infty$.
- **De Broglie Wavelength:** The de Broglie wavelength of an orbital electron in state n is $\lambda = \frac{2\pi r_n}{n}$. Since $r_n \propto n^2$, the de Broglie wavelength scales linearly with n: $\lambda \propto n$.

Final Exam-Day Checklist

- ✓ **Geiger-Marsden Law:** $N(\theta) \propto \frac{1}{\sin^4(\theta/2)}$ and $b \propto \cot(\theta/2)$.
- ✓ **Closest Approach:** $d = \frac{2Ze^2}{4\pi\varepsilon_0 K} \propto \frac{Z}{K}$.
- ✓ **Bohr Scalings:** $r_n = 0.529\frac{n^2}{Z}$ Å, $v_n = 2.18 \times 10^6 \frac{Z}{n}$ m/s, $E_n = -13.6\frac{Z^2}{n^2}$ eV.
- ✓ **Frequency and Period:** $f_n \propto \frac{Z^2}{n^3}$ and $T_n \propto \frac{n^3}{Z^2}$.
- ✓ **De Broglie Standing Wave:** $2\pi r_n = n\lambda \implies mvr = \frac{nh}{2\pi}$.
- ✓ **Rydberg Formula:** $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$ with $\frac{1}{R} \approx 912$ Å.

BOSONANCE

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